PADHAKU
Curiosity · Class 6 Science



CHAPTER 6

Materials Around Us

NCERT Exercise Solutions

6.1 Observing Objects Around Us	1 Q
6.2 How to Group Materials?	12 Q
6.3.1 Observe and identify appearance of materials	2 Q
6.3.2 Which materials are hard?	3 Q
6.3.3 Explore materials through which one can see or cannot see	5 Q
6.3.4 What is soluble in water; what is not?	4 Q
6.3.5 How heavy or light?	1 Q
6.3.6 Space and volume	3 Q
6.4 What is Matter?	4 Q
Let us play	2 Q
Let us enhance our learning	11 Q
Learning further	4 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

Independent study aid by Beginers Point. Not an official NCERT publication.

⭐ 6.1 Observing Objects Around Us

Activity 6.1 **Make a list of objects you see around and write the names of the materials they are made up of (Table 6.1).**

💡 **SAMPLE ANSWER**

I observe	Materials they are made up of
Notebook	Paper
Pen	Plastic, metal, ink
Table	Wood, iron
Window	Glass, wood or iron
Bucket	Plastic or iron
Shirt	Cloth (cotton)

Conclusion: Objects around us are made up of various materials.

⭐ 6.2 How to Group Materials?

Activity 6.2 **Group the objects shown in Fig. 6.1 based on any common property, such as shape, colour, hardness, softness, shine, dullness or materials they are made up of.**

💡 **SAMPLE ANSWER**

We can group them, for example, by material:

- Metal objects (shiny): spoon, key, coins
- Plastic objects: bottle, bucket, toys
- Wooden objects: chair, pencil

- Cloth/soft objects: towel, pillow

We can also group them by shape (round / flat) or by hard / soft. (check the picture in your book)

? Which property did you use to group the objects in Activity 6.2?

💡 **SAMPLE ANSWER**

I grouped them by the material they are made of (metal, plastic, wood, cloth). Shape, colour, or hard/soft can also be used.

? Did your friends group objects based on similar properties?

💡 **SAMPLE ANSWER**

Some friends used the same property; others used different ones, like shape or colour. So the same objects can be grouped in many ways.

? What did you learn from this activity?

✔ **ANSWER**

Objects can be grouped (classified) on the basis of a common property.

An object can be made from different materials, and one material can be used to make many objects.

Activity 6.3 **What materials can we use to make a tumbler? Fill in the names of the materials (Fig. 6.2).**

✔ ANSWER

Glass, steel (metal), plastic, copper, brass, clay (earthen tumbler/kulhad).

These materials can hold water.

? **Would it be a good idea to use paper-like materials for making cooking utensils?**

✔ ANSWER

No. Paper burns easily, gets soaked and torn by water, and is not strong. Cooking utensils must bear heat and hold food and water.

? **Why cannot a tumbler made of cloth be used for storing water?**

✔ ANSWER

Cloth has tiny holes and soaks up water, so water would leak out. A tumbler must be made of a material that can hold water.

? **Can a cricket ball be used to play tennis?**

✔ ANSWER

No. A cricket ball is hard and heavy and does not bounce much. Tennis needs a light ball that bounces high.

? **Why are different materials used for making balls for various sports?**

✔ ANSWER

Each sport needs a ball with special properties (hardness, weight, bounce, size). So the material is chosen to suit the game.

Activity 6.4 **Drop each ball (tennis ball, cricket ball, hand exercise ball) from a fixed height. Note how high it bounces (Table 6.2). Which ball bounces highest?**

💡 SAMPLE ANSWER

Ball	Bounce (high, medium or low)
Tennis ball	High
Cricket ball	Low
Hand exercise ball	Medium

The **tennis ball** bounces the highest. Balls are made of specific materials to suit specific sports.

? **Observe Fig. 6.4 and group the objects in as many different ways as possible.**

💡 SAMPLE ANSWER

- By shape: round, flat, long objects
- By colour: red, blue, white... objects
- By material: metal, plastic, wood, glass, cloth
- By hardness: hard and soft objects

(check the picture in your book)

? Visit a chemist shop and enquire about how medicines are arranged.

💡 SAMPLE ANSWER

Medicines are kept in groups: tablets, syrups, ointments and injections separately, often in alphabetical order, and some (like insulin) in a fridge. This helps find them quickly.

⭐ 6.3.1 Observe and identify appearance of materials

? Sorting challenge: Collect pieces of paper, cardboard, wood, chalk, copper wire, aluminium foil, brass, bronze, steel. Do any of these shine when light falls on them? Group them based on appearance.

✔ ANSWER

Shiny (lustrous)	Not shiny (non-lustrous)
Copper wire, aluminium foil, brass, bronze, steel	Paper, cardboard, wood, chalk

Shiny materials are mostly metals.

? **Are all lustrous materials metals?**

✔ ANSWER

No. "All that glitters is not gold." Some materials are made shiny by polishing or by coating them with plastic or wax. They may not be metals.

⭐ 6.3.2 Which materials are hard?

? **Scratch wood, aluminium, stone, iron, candle, chalk, etc. with a metal key. Can some materials be scratched more easily than others?**

✔ ANSWER

Yes. Candle, chalk and wood get scratched easily (soft). Stone and iron are hard to scratch (hard).

Activity 6.5 **Hold the objects and feel whether they are hard or soft. Find out the materials they are made up of (Table 6.3).**

✔ ANSWER

Object	Hard/Soft	Material(s)
Brick	Hard	Baked clay
Water bottle	Hard	Plastic or steel
Pillow	Soft	Cotton, cloth
Tumbler	Hard	Glass or steel

Table	Hard	Wood or iron
Sweater	Soft	Wool

? **Can you think of any other properties that are shown by materials?**

💡 **SAMPLE ANSWER**

Yes: whether we can see through them, whether they dissolve in water, whether they are heavy or light, and whether they float or sink.

⭐ 6.3.3 Explore materials through which one can see or cannot see

? **Why did Ghulan, Sheeta and Sara choose these places to hide?**

✔ **ANSWER**

We cannot see through a wall or a big tree (opaque), so Ghulan and Sheeta could not be seen. Through the frosted glass door Sara could be seen only unclearly (translucent).

? **Would it be possible for Sheeta's brother to see her and her friends through a closed wooden window?**

✔ **ANSWER**

No. Wood is opaque, so we cannot see through it.

Fig. 6.5 **Identify the nature of materials used by Ghulan (A), Sheeta (B), Sara (C) and Sheeta's brother (D).**

✔ ANSWER

(A) Wall: **opaque**

(B) Tree: **opaque**

(C) Frosted glass door: **translucent**

(D) Glass window: **transparent**

Activity 6.6 **Classify as transparent, translucent or opaque: glass tumbler, butter paper, eraser, frosted glass, wooden board, window glass (Table 6.4).**

✔ ANSWER

Transparent	Translucent	Opaque
Glass tumbler	Butter paper	Eraser
Window glass	Frosted glass	Wooden board

? **Is water transparent? Can it be made opaque?**

✔ ANSWER

Yes, clean water is transparent.

Yes, it can be made opaque by mixing a lot of mud, ink or colour in it; then we cannot see through it.

⭐ 6.3.4 What is soluble in water; what is not?

Activity 6.7 **Mix sugar, salt, chalk powder, sand and sawdust in five glass tumblers of water and stir. Predict and observe (Table 6.5).**

✔ ANSWER

Material	Observation
Sugar	Disappears in water
Salt	Disappears in water
Chalk powder	Does not disappear (water turns milky, powder settles)
Sand	Does not disappear (settles at the bottom)
Sawdust	Does not disappear (floats on top)

Sugar and salt are **soluble**; chalk powder, sand and sawdust are **insoluble** in water.

? **Does everything you put in water disappear?**

✔ ANSWER

No. Only soluble materials like sugar and salt disappear. Insoluble materials like sand, chalk powder and sawdust do not.

? **Do liquids like oil, vinegar and honey dissolve in water? Explore.**

✔ ANSWER

Vinegar and honey mix completely with water.

Oil does not mix; it forms a separate layer on top of water.

? **What about the gases present in water?**

✔ ANSWER

Some gases dissolve in water and some do not. Oxygen dissolves in water; animals and plants living in water need it to survive.

⭐ 6.3.5 How heavy or light?

Activity 6.8 **Fill three identical cups half with water (A), sand (B) and pebbles (C). Predict which is heavier and which is lighter, then weigh them on a balance and compare.**

💡 SAMPLE ANSWER

The cup with water (A) is usually the lightest; the cups with sand (B) and pebbles (C) are heavier. Check with the balance.

Conclusion: The heavier object has more mass; the lighter one has less mass.

⭐ 6.3.6 Space and volume

? Why is the level of water different in the two tumblers?

✔ ANSWER

Both tumblers have the same capacity, but the amount of water poured is different. Less water occupies less space, so its volume and level are less.

? Why was I not able to transfer water from the jug into an empty water bottle completely?

✔ ANSWER

The jug had more water than the bottle can hold. The bottle's volume (space inside) is less than the volume of water in the jug.

? Have you noticed 1 L, 500 mL, 200 mL, etc., written as net quantity on bottles? What do these indicate?

✔ ANSWER

Yes. They tell the **volume** of water (or milk) in the bottle.

⭐ 6.4 What is Matter?

? Are there any properties which can be shown by all materials? If yes, what are those?

✔ ANSWER

Yes. **Mass** and **volume**: all materials have mass and occupy space.

? Is air matter?

✔ ANSWER

Yes. Air has mass and occupies space (it fills a balloon), so air is matter.

? Can all the materials around us be considered as different examples of matter? Discuss.

✔ ANSWER

Yes. Every material (water, sand, pebbles, the cup) has mass and occupies space, so all materials are matter.

Think it over! What changes did the invention of plastic bring to humans? Is it a boon or a bane?

💡 SAMPLE ANSWER

Boon: plastic is light, strong, cheap, waterproof and easy to shape, so it is used for bottles, buckets, pipes, toys and medical items.

Bane: it does not rot easily, so plastic waste pollutes land and water and harms animals.

So plastic is useful, but we must use less of it and recycle it.

Let us play

1 **Find the companion. Link words that have a connection:**

Transparent, Solid, Plastic, Wood, Glass

Iron, Copper, Bottle, Lustrous, Opaque

💡 SAMPLE ANSWER

- Glass → Transparent
- Iron → Lustrous, Solid
- Copper → Lustrous
- Plastic → Bottle
- Wood → Opaque

(Other correct links are also possible.)

2 **Win the 'Word-hub': choose nine words from the list (lustrous, non-lustrous, soluble, insoluble, hard, soft, matter, mass, transparent, opaque, volume, translucent) and tick them when the facilitator reads the word or its definition.**

💡 SAMPLE ANSWER

Meanings to remember:

- Lustrous: shiny surface; Non-lustrous: not shiny
- Soluble: mixes completely in water; Insoluble: does not
- Hard: difficult to compress or scratch; Soft: easy to compress or scratch
- Matter: occupies space and has mass
- Mass: amount of matter; Volume: space occupied
- Transparent: see clearly through; Translucent: see, but not clearly; Opaque: cannot see through

Let us enhance our learning

1 **Visit your kitchen and observe how your parents have organised various edibles. Can you suggest a better sorting method?**

💡 SAMPLE ANSWER

Edibles are kept in groups: pulses together, spices together, grains together.

Better: keep them in labelled transparent jars, items used daily in front, and things like milk and vegetables in a cool place.

2 **Unscramble the letters (Column I) and match with their properties (Column II).**

Column I	Column II

(i) T R E M A T	(a) Objects can be seen clearly through it
(ii) U L S B E L O	(b) Occupies space and has mass
(iii) T N E R P A S N A R T	(c) Shiny surface
(iv) E R U S T L	(d) Mixes completely in water

✔ ANSWER

(i) **MATTER** → (b)

(ii) **SOLUBLE** → (d)

(iii) **TRANSPARENT** → (a)

(iv) **LUSTRE** → (c)

3 **The containers which are used to store materials in shops and at home are usually transparent. Give your reasons for this.**

✔ ANSWER

- We can see what is inside without opening them.
- We can see how much is left, so we know when to refill.
- We can check if the food is clean and fresh.

4 **State True [T] or False [F]. Correct the false statements.**

(i) Wood is translucent while glass is opaque.

(ii) Aluminium foil has lustre while an eraser does not.

(iii) Sugar dissolves in water whereas sawdust does not.

(iv) An apple is a matter because it occupies no space and has mass.

✔ ANSWER

(i) **F**. Wood is opaque while glass is transparent.

(ii) **T**

(iii) **T**

(iv) **F**. An apple is matter because it occupies space and has mass.

5 **Chairs are made of wood, iron, plastic, bamboo, cement and stones. Which materials fulfil these properties the most?**

(i) Hardness (does not bend or shake even after long use)

(ii) Lightweight (easy to lift or move)

(iii) Does not feel very cold in winters

(iv) Can be cleaned regularly and made to look new after long use

💡 SAMPLE ANSWER

(i) Iron, stone, cement, wood

(ii) Plastic, bamboo

(iii) Wood, bamboo, plastic

(iv) Plastic, iron (can be repainted), stone

6 **You need containers for collecting (i) food waste, (ii) broken glass and (iii) wastepaper. Which materials will you choose? What properties do you need to think of?**

💡 **SAMPLE ANSWER**

(i) Food waste: plastic or metal bin with a lid; it should not leak (waterproof) and be easy to wash.

(ii) Broken glass: strong metal or thick hard plastic container; it should be hard so glass cannot cut or break it.

(iii) Wastepaper: cardboard box, cane/bamboo basket or cloth bag; it can be light and cheap.

Properties: hardness/strength, waterproof, easy to clean, weight and cost.

7 **Air does not hinder us from seeing each other, but a wooden door does. It is because air is ____ and the wooden door is ____.**

(i) transparent, opaque

(ii) translucent, transparent

(iii) opaque, translucent

(iv) transparent, translucent

✔ **ANSWER**

(i) transparent, opaque. We can see clearly through air but not at all through wood.

8 **Material X is rigid and does not change shape when pressed; Y changes shape easily. In water, only X dissolves completely, Y remains unchanged. What can X and Y be? Is X hard or soft? What about Y? Justify.**

💡 SAMPLE ANSWER

X can be a lump of sugar candy (mishri) or a crystal of rock salt. X is **hard**, because it does not change shape when pressed, and it is soluble.

Y can be a sponge or a rubber eraser. Y is **soft**, because it changes shape easily when pressed, and it is insoluble.

9 (i) **Who am I? Identify me on the basis of the given properties.**

(a) I have lustre.

(b) I can be easily compressed.

(c) I am hard and soluble in water.

(d) You cannot see clearly through me.

(e) I have mass and volume but you cannot see me.

✔ ANSWER

(a) A metal, e.g. gold, iron or aluminium foil

(b) Sponge, cotton or a pillow

(c) Sugar candy (mishri) or a salt crystal

(d) A translucent material, e.g. butter paper or frosted glass

(e) Air

9 (ii) **Make your own 'Who am I?'**

💡 SAMPLE ANSWER

I am transparent, I am a liquid, and I can dissolve sugar and salt. Who am I? **Water**

I am lustrous, I am used to make wires, and I am reddish-brown. Who am I? **Copper**

10 **Materials: vinegar, honey, mustard oil, water, glucose and wheat flour. Make two pairs where one material is soluble in the other, and two pairs where one remains insoluble in the other.**

✔ ANSWER

Soluble pairs: glucose in water; vinegar in water (honey in water is also correct).

Insoluble pairs: mustard oil in water; wheat flour in water.

📘 Learning further

1 **Gather information on different materials which can be recycled.**

💡 SAMPLE ANSWER

Paper and cardboard, many plastics, glass, metals (iron, aluminium, copper, brass) and cloth can be recycled.

2 **Survey recyclers near you: what properties of materials do they check before buying objects? Which materials do they not buy and why?**

💡 **SAMPLE ANSWER**

They check what the material is (metal, plastic, paper, glass), whether it is clean and dry, and its weight. They may test a metal with a magnet.

They usually do not buy dirty, wet or mixed materials (like plastic-coated paper or thermocol) because these are hard to recycle and give little money.

3 **Collect 20–30 objects from your household and classify them based on the properties of their materials. What relationship do you see between the properties of the materials and the use of the objects?**

💡 **SAMPLE ANSWER**

Example groups: hard/soft, lustrous/non-lustrous, transparent/translucent/opaque.

The use of an object depends on the properties of its material: a cooking pot is metal (bears heat), a window is glass (transparent), a pillow is cotton (soft).

4 **Create and decorate a useful object using discarded materials and bring it to class. Discuss and give constructive feedback.**

💡 **SAMPLE ANSWER**

Example: a pen stand made from an empty plastic bottle, decorated with coloured paper. Plastic is light and strong, so it holds pens well. Feedback: make the base heavier so it does not fall over.

🎉 **Chapter 6 done! Revise the answers once more tomorrow.**